TOSHIBA Bi-CD Integrated Circuit Silicon Monolithic
TB6612FNG
Driver IC for Dual DC Motor

1. Description & Features
The TB6612FNG is a driver IC for dual DC motor with output transistor in LD MOS structure with low ON-resistor.
Two input signals, IN1 and IN2, can choose one of four modes such as CW, CCW, short brake, and stop mode.

Key Features:
- Power supply voltage: Large motor drive capability (VM = 15 V max)
- Output current: IOUT = 1.2 A (average) / 3.2 A (peak pulse < 20 ms)
- Output low ON resistance: Ron = 0.5 ohm (upper + lower typ. at VM = 5V)
- Standby (power save) system: STBY pin (Active HIGH)
- Built-in thermal shutdown (TSD) circuit
- Built-in low voltage detection circuit (UVLO: VCC < 2.2V typ)
- Small package: SSOP24 (0.65 mm pin pitch)

Absolute Maximum Ratings (Ta = 25 deg C):
- Supply voltage (VM): 15.0 V
- Control supply voltage (VCC): 6.0 V
- Output current (IOUT peak): 3.2 A
- Operating temperature (Topr): -20 to 85 deg C
- Storage temperature (Tstg): -55 to 150 deg C

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2. Pin Functions
- Pin 24 (VM1) & Pin 13 (VM2, VM3): Motor supply voltage pins (4.5 V to 13.5 V recommended).
- Pin 20 (VCC): Control logic supply voltage pin (2.7 V to 5.5 V).
- Pin 19 (STBY): Standby input pin. HIGH = Normal operation, LOW = Standby (all outputs Hi-Z).
- Pin 21 (PWMA) & Pin 16 (PWMB): PWM speed control input (up to 100 kHz frequency).
- Pin 22 (AIN1) & Pin 23 (AIN2): Motor A direction logic control pins.
- Pin 17 (BIN1) & Pin 18 (BIN2): Motor B direction logic control pins.
- Pin 1, 2 (AO1) & Pin 5, 6 (AO2): Motor A driver outputs.
- Pin 7, 8 (BO2) & Pin 11, 12 (BO1): Motor B driver outputs.
- Pin 3, 4, 9, 10, 14, 15 (PGND, SGND): Ground connection pins. Must connect to system GND.

3. Recommended Operating Conditions (Ta = -20 to 85 deg C):
- Supply voltage (VM): 4.5 V to 13.5 V
- Logic supply voltage (VCC): 2.7 V to 5.5 V
- Output current (IOUT continuous): 1.0 A (per channel)
- PWM frequency (fPWM): Up to 100 kHz

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4. Input/Output Control Function Truth Table

| IN1 | IN2 | PWM | STBY | OUT1 | OUT2 | Mode |
|---|---|---|---|---|---|---|
| H | L | H | H | H | L | Clockwise (CW) |
| L | H | H | H | L | H | Counter-Clockwise (CCW) |
| H | H | X | H | L | L | Short Brake |
| L | L | X | H | OFF | OFF | Stop (Hi-Z coast) |
| X | X | X | L | OFF | OFF | Standby Mode (Outputs Hi-Z, low current) |

5. Thermal Shutdown and Circuit Protection

- Thermal shutdown operates at junction temperature Tj >= 175 deg C (typ).
- Hysteresis temperature is approximately 40 deg C.
- Bypass capacitor: Connect a 0.1 uF ceramic capacitor and a 10 uF electrolytic capacitor between VM and GND, and between VCC and GND, as close as possible to the IC pins.
- Flyback protection: Internal parasitic diodes clamp inductive flyback from motor coils.